

Kari Stadius

senior researcher

基本信息

姓名:	Kari Stadius	所属院校:	阿尔托大学
地区:	Finland	年龄范围:	30岁至40岁

学科领域

通信工程 微电子科学与工程 集成电路设计与集成系统

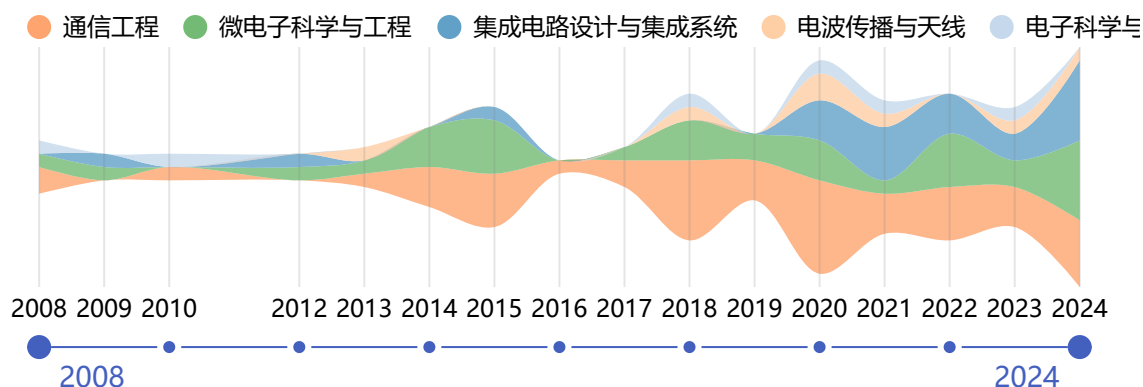
研究领域

射频集成电路 无线通信与智能天线 移动通信与5G/6G 模拟与数模混合电路
模拟与射频IC设计 通信系统仿真与优化

个人简介

Kari Stadius is a Staff Scientist in the Department of Electronics and Nanoengineering and the Jussi Ryyänen Group. With an ORCID of 0000 - 0001 - 8775 - 9287 and phone +358 50 3218825, based at Maarintie 8. They hold a Doctoral degree from Aalto University (2010), a Licentiate degree from Helsinki University of Technology (1997), and a Master's degree from the same institution (1994). Their research focuses on areas like Receivers, Transmitter, and Design. Kari has 169 research outputs including conference articles, papers, and patents, and is involved in 12 projects such as JOLI and RF Sampo.

近年发表的论文



Reconfigurable Signal Processing and DSP Hardware Generator for 5G and Beyond Transmitters

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ghosh, Agnimesh; Spelman, Andrei; Cheung, Tze Hin; Boopathy, Dhanashree; Stadius, Kari; Gomony, Manil Dev; Valkama, Mikko; Ryyanen, Jussi; Kosunen, Marko; Unnikrishnan, Vishnu

被引用次数: 1

2024

55-100-GHz Enhanced Gilbert Cell Mixer Design in 22-nm FDSOI CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Jokiniemi, Kimi; Ryyanen, Kaisa; Vaha, Joni; Kankkunen, Elmo; Stadius, Kari; Ryyanen, Jussi

被引用次数: 0

2024

An 18-28 GHz dual-mode down-converter IC for 5G applications

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Naghavi, Saeed; Ryyanen, Kaisa; Zahra, Mahwish; Korsman, Aleks; Stadius, Kari; Kosunen, Marko; Unnikrishnan, Vishnu; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 0

2024

Variable Time-Step 2.5-15 GHz Phase Modulator with Pre-Distortion for Outphasing Transmitters

2024 22ND IEEE INTERREGIONAL NEWCAS CONFERENCE, NEWCAS 2024

Cheung, Tze Hin; Ghosh, Agnimesh; Spelman, Andrei; Boopathy, Dhanashree; Unnikrishriant, Vishnu; Anttila, Lauri; Valkamat, Mikko; Kosunen, Marko; Stadius, Kari; Ryyanlinen, Jussi

被引用次数: 0

2024

A Compact and Wideband mmWave Passive CMOS Circulator Based on Switched All-Pass Networks

IEEE MICROWAVE AND WIRELESS TECHNOLOGY LETTERS

Naghavi, Saeed; Gao, Jian; Stadius, Kari; Ryyanen, Jussi

被引用次数: 1

2024

Crack Stop as a Coupling Element Between an IC Chip and Antenna

2024 18TH EUROPEAN CONFERENCE ON ANTENNAS AND PROPAGATION, EUCAP

Bergman, Jan H. S.; Ryyanen, Kaisa; Ala-Laurinaho, Juha; Stadius, Kari; Ryyanen, Jussi; Viikari, Ville

被引用次数: 0

2024

Analysis of Current-Commutating Passive and Active Mixers for mmWave Applications

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Jokiniemi, Kimi; Ryyananen, Kaisa; Vaha, Joni; Stadius, Kari; Ryyananen, Jussi

被引用次数: 0

2024

Analysis and Design of Constant-Slope Voltage-to-Time Converters

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Porrasmaa, Santeri; Jarvinen, Okko; Kempfi, Ilia; Stadius, Kari; Kosunen, Marko; Ryyananen, Jussi

被引用次数: 0

2024

A Wideband IF Receiver Chip for Flexibly Scalable mmWave Subarray Combining and Interference Rejection

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Akbar, Rehman; Klumperink, Eric A. M.; Tervo, Nuutti; Stadius, Kari; Rahkonen, Timo; Parssinen, Aarno

被引用次数: 2

2023

An Automated EM-Simulation Environment with Parameterized Layout Generation for Microwave Integrated Circuits

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Ryyananen, Kaisa; Lahtinen, Veeti; Porrasmaa, Santeri; Stadius, Kari; Kosunen, Marko; Ryyananen, Jussi

被引用次数: 0

2023

Active Wideband 55-100-GHz Downconversion Mixer in 22-nm FDSOI CMOS

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Jokiniemi, Kimi; Ryyananen, Kaisa; Vaha, Joni; Kankkunen, Elmo; Stadius, Kari; Ryyananen, Jussi

被引用次数: 0

2023

Leveraging frequency agility of an MIMO antenna cluster with a transmitter IC

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Saleem, Ali Raza; Luomaniemi, Rasmus; Lehtovuori, Anu; Stadius, Kari; Kosunen, Marko; Viikari, Ville; Ryyananen, Jussi

被引用次数: 0

2023

True-Time-Delay Receiver IC With Reconfigurable Analog and Digital Beamforming

IEEE ACCESS

Spooft, Kalle; Tenhunen, Miikka; Unnikrishnan, Vishnu; Stadius, Kari; Kosunen, Marko; Ryyananen, Jussi

被引用次数: 3

2022

Energy-Efficient Cyclic-Coupled Ring Oscillator With Delay-Based Injection Locking

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jarvinen, Okko; Unnikrishnan, Vishnu; Kempfi, Ilia; Porrasmaa, Santeri; Stadius, Kari; Kosunen, Marko; Ryyananen, Jussi

被引用次数: 2

2022

A 0.9-Nyquist-Band Digital Timing Mismatch Correction for Time-Interleaved ADCs Achieving Delay Tuning Range of 0.12-Sample-Period

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Kempi, Ilia; Jarvinen, Okko; Kosunen, Marko; Unnikrishnan, Vishnu; Stadius, Kari; Ryyanen, Jussi

被引用次数: 1

2022

Reconfigurable Signal Processing and DSP Hardware Generator for 5G Transmitters

2022 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE (NORCAS)

Ghosh, Agnimesh; Spelman, Andrei; Cheung, Tze Hin; Boopathy, Dhanashree; Unnikrishnan, Vishnu; Lampu, Vesa; Xu, Guixian; Anttila, Lauri; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 3

2022

Design of Cyclic-Coupled Ring Oscillators with Guaranteed Maximal Phase Resolution

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Jarvinen, Okko; Unnikrishnan, Vishnu; Kemp, Ilia; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 0

2022

A 0.8-6 GHz True-Time-Delay Beam-Nulling Receiver Front-End

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Spoo, Kalle; Tenhunen, Miikka; Unnikrishnan, Vishnu; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 0

2022

A 6-20 GHz 400-MHz Modulation-Bandwidth CMOS Transmitter IC

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Saleem, Ali Raza; Naghavi, Saeed; Zahra, Mahwish; Stadius, Kari; Kosunen, Marko; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 2

2022

Data Conversion With Subgate-Delay Time Resolution Using Cyclic-Coupled Ring Oscillators

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Unnikrishnan, Vishnu; Jarvinen, Okko; Siddiqui, Waqas; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 5

2021

Characterization of an Antenna Cluster and Transmitter IC with a Modulated Signal

2021 IEEE MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM (IMS)

Saleem, Ali Raza; Stadius, Kari; Hannula, Jari-Matti; Lehtovuori, Anu; Kosunen, Marko; Viikari, Ville; Ryyanen, Jussi

被引用次数: 0

2021

Quantized Polar Transmitters for Power Efficient Massive MIMO Systems

IEEE WIRELESS COMMUNICATIONS LETTERS

Xu, Guixian; Lampu, Vesa; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko; Anttila, Lauri

被引用次数: 1

2021

A Transmitter IC with Supply Tuning for Frequency-Reconfigurable Antenna Cluster

2021 IEEE INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS (MWSCAS)

Saleem, Ali Raza; Stadius, Kari; Hannula, Jari-Matti; Lehtovuori, Anu; Kosunen, Marko; Viikari, Ville; Ryyanen, Jussi

被引用次数: 0

2021

A 38.5-to-60.5 GHz LNA with Wideband Combiner Supporting Cartesian Beamforming Architecture

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Akbar, Rehman; Shaheen, Rana A.; Rahkonen, Timo; Tze, Cheung; Stadius, Kari; Parssinen, Aarno

被引用次数: 3

2021

A 100-750 MS/s 11-Bit Time-to-Digital Converter With Cyclic-Coupled Ring Oscillator

IEEE ACCESS

Jarvinen, Okko; Unnikrishnan, Vishnu; Siddiqui, Waqas; Korhonen, Teuvo; Koli, Kimmo; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 9

2021

True-Time-Delay Beamforming Receiver With RF Re-Sampling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Spoof, Kalle; Unnikrishnan, Vishnu; Zahra, Mahwish; Stadius, Kari; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 20

2020

A Class-D Tri-Phasing CMOS Power Amplifier With an Extended Marchand-Balun Power Combiner

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Martelius, Mikko; Stadius, Kari; Lemberg, Jerry; Roverato, Enrico; Nieminen, Tero; Antonov, Yury; Anttila, Lauri; Valkama, Mikko; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 5

2020

A 2-5.5 GHz Beamsteering Receiver IC With 4-Element Vivaldi Antenna Array

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Zahra, Mahwish; Kemp, Ilia; Haarla, Jaakko; Antonov, Yury; Khonsari, Zahra; Miilunpalo, Toni; Ahmed, Nouman; Inkinen, Juha; Unnikrishnan, Vishnu; Lehtovuori, Anu; Viikari, Ville; Anttila, Lauri; Valkama, Mikko; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi

被引用次数: 4

2020

Passive Intermodulation in Simultaneous Transmit-Receive Systems: Modeling and Digital Cancellation Methods

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Waheed, Muhammad Zeeshan; Korpi, Dani; Anttila, Lauri; Kiayani, Adnan; Kosunen, Marko; Stadius, Kari; Campo, Pablo Pascual; Turunen, Matias; Allen, Markus; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 27

2020

A 1.5-5-GHz Integrated RF Transmitter Front End for Active Matching of an Antenna Cluster

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Saleem, Ali Raza; Stadius, Kari; Hannula, Jari-Matti; Lehtovuori, Anu; Kosunen, Marko; Viikari, Ville; Ryyanen, Jussi

A Six-Phase Two-Stage Blocker-Tolerant Harmonic-Rejection Receiver

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ul Haq, Faizan; Englund, Mikko; Antonov, Yury; Tenhunen, Miikka; Stadius, Kari; Kosunen, Marko; Ostman, Kim B.; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 4

2020

A wideband blocker-resilient direct delta sigma receiver with selective input-impedance matching

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Ul Haq, Faizan; Englund, Mikko; ostman, Kim B.; Stadius, Kari; Kosunen, Marko; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 0

2020

Full-Duplex OFDM Radar With LTE and 5G NR Waveforms: Challenges, Solutions, and Measurements

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Barneto, Carlos Baquero; Riihonen, Taneli; Turunen, Matias; Anttila, Lauri; Fleischer, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 187

2019

Quantization noise upconversion effects in mixer-first direct delta-sigma receivers

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Ul Haq, Faizan; Englund, Mikko; Ostman, Kim B.; Stadius, Kari; Kosunen, Marko; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 0

2019

A 1.5-1.9-GHz All-Digital Tri-Phasing Transmitter With an Integrated Multilevel Class-D Power Amplifier Achieving 100-MHz RF Bandwidth

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lemberg, Jerry; Martelius, Mikko; Roverato, Enrico; Antonov, Yury; Nieminen, Tero; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Kosunen, Marko; Ryyanen, Jussi

被引用次数: 24

2019

A common-gate common-source low noise amplifier based RF front end with selective input impedance matching for blocker-resilient receivers

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Ul Haq, Faizan; Ostman, Kim B.; Englund, Mikko; Stadius, Kari; Kosunen, Marko; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 4

2018

A Systematic Design Method for Direct Delta-Sigma Receivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Englund, Mikko; Ul Haq, Faizan; Stadius, Kari; Kosunen, Marko; Ostman, Kim B.; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 2

2018

A High-Speed DSP Engine for First-Order Hold Digital Phase Modulation in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Roverato, Enrico; Kosunen, Marko; Lemberg, Jerry; Martelius, Mikko; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 3

2018

Performance Analysis of Frequency-Reconfigurable Antenna Cluster With Integrated Radio Transceivers

IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS

Hannula, Jari-Matti; Kosunen, Marko; Lehtovuori, Anu; Rasilainen, Kimmo; Stadius, Kari; Ryyanen, Jussi; Viikari, Ville

被引用次数: 13

2018

Adaptive Nonlinear RF Cancellation for Improved Isolation in Simultaneous Transmit-Receive Systems

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kiayani, Adnan; Waheed, Muhammad Zeeshan; Anttila, Lauri; Abdelaziz, Mahmoud; Korpi, Dani; Syrjala, Ville; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 72

2018

Tri-Phasing Modulation for Efficient and Wideband Radio Transmitters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lemberg, Jerry; Martelius, Mikko; Kosunen, Marko; Roverato, Enrico; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 4

2018

Modeling and Joint Mitigation of TX and RX Nonlinearity-Induced Receiver Desensitization

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kiayani, Adnan; Anttila, Lauri; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 16

2017

Reference Receiver Enhanced Digital Linearization of Wideband Direct-Conversion Receivers

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Marttila, Jaakko; Allen, Markus; Kosunen, Marko; Stadius, Kari; Ryyanen, Jussi; Valkama, Mikko

被引用次数: 18

2017

Digital Interpolating Phase Modulator for Wideband Outphasing Transmitters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lemberg, Jerry; Kosunen, Marko; Roverato, Enrico; Martelius, Mikko; Stadius, Kari; Anttila, Lauri; Valkama, Mikko; Ryyanen, Jussi

被引用次数: 26

2016

Next-Generation RF Front-End Design Methods for Direct $\Delta\Sigma$ Receivers

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Oestman, Kim B.; Englund, Mikko; Viitala, Olli; Stadius, Kari; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 6

2015

A Programmable 0.7-2.7 GHz Direct $\Delta\Sigma$ Receiver in 40 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Englund, Mikko; Ostman, Kim B.; Viitala, Olli; Kaltiokallio, Mikko; Stadius, Kari; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 31

2015

A 5.8-Gbps low-noise scalable low-voltage signaling serial link transmitter for MIPI M-PHY in 40-nm CMOS

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Nieminen, Tero; Tikka, Tero; Antonov, Yury; Viitala, Olli; Stadius, Kari; Voutilainen, Martti; Ryyanen, Jussi

被引用次数: 4

2015

Analysis and Design of N-Path Filter Offset Tuning in a 0.7-2.7-GHz Receiver Front-End

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ostman, Kim B.; Englund, Mikko; Viitala, Olli; Kaltiokallio, Mikko; Stadius, Kari; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 23

2015

RF Input Matching Design for Closed-Loop Direct Delta-Sigma Receivers

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ostman, Kim B.; Englund, Mikko; Viitala, Olli; Stadius, Kari; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 2

2015

Characteristics of LNA Operation in Direct Delta-Sigma Receivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ostman, Kim B.; Englund, Mikko; Viitala, Olli; Stadius, Kari; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 9

2014

A 2.5-GHz Receiver Front-End With Q-Boosted Post-LNA N-Path Filtering in 40-nm CMOS

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ostman, Kim B.; Englund, Mikko; Viitala, Olli; Kaltiokallio, Mikko; Stadius, Kari; Koli, Kimmo; Ryyanen, Jussi

被引用次数: 13

2014

RX-Band Noise Reduction in All-Digital Transmitters With Configurable Spectral Shaping of Quantization and Mismatch Errors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Roverato, Enrico; Kosunen, Marko; Lemberg, Jerry; Stadius, Kari; Ryyanen, Jussi

被引用次数: 8

2014

A 0.7-2.7-GHz Blocker-Tolerant Compact-Size Single-Antenna Receiver for Wideband Mobile Applications

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kaltiokallio, Mikko; Valkonen, Risto; Stadius, Kari; Ryyanen, Jussi

An All-Digital PLL Frequency Synthesizer With an Improved Phase Digitization Approach and an Optimized Frequency Calibration Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xu, Liangge; Stadius, Kari; Ryyanen, Jussi

被引用次数: 13

2012

A 2.4-GHz Low-Power All-Digital Phase-Locked Loop

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Liangge; Lindfors, Saska; Stadius, Kari; Ryyanen, Jussi

被引用次数: 46

2010

A Digital Frequency Synthesizer for Cognitive Radio Spectrum Sensing Applications

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Rapinoja, Tapio; Stadius, Kari; Xu, Liangge; Lindfors, Saska; Kaunisto, Risto; Parssinen, Aarno; Ryyanen, Jussi

被引用次数: 17

2010

A digitally controlled 2.4-GHz oscillator in 65-nm CMOS

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Xu, Liangge; Lindfors, Saska; Stadius, Kari; Ryyanen, Jussi

被引用次数: 5

2009

Analysis and Design of Passive Polyphase Filters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kaukovaari, Jouni; Stadius, Kari; Ryyanen, Jussi; Halonen, Kari A. I.

被引用次数: 152

2008

A low-power phase-locked loop for UWB applications

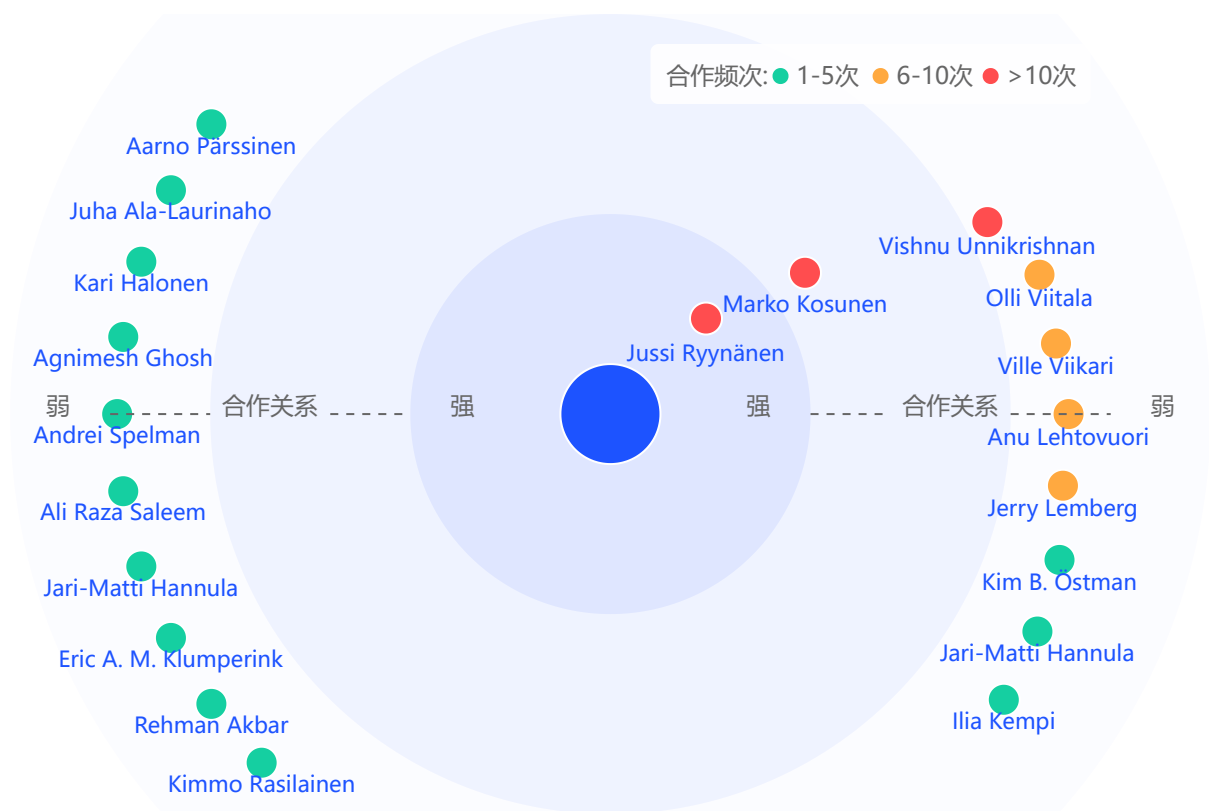
ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Rapinoja, Tapio; Stadius, Kari; Halonen, Kari

被引用次数: 2

2008

学术合作



Jussi Ryyänen

阿尔托大学

合作次数: 52

Marko Kosunen

阿尔托大学

合作次数: 37

Vishnu Unnikrishnan

阿尔托大学

合作次数: 11

Olli Viitala

阿尔托大学

合作次数: 7

Ville Viikari

阿尔托大学

合作次数: 7

还有 16 个合作者